Benjamin Andrew

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Education

University of Manchester, PhD in Computer Science — Sep 2024 – Present

- Project titled *Developing Reliable Software for Autonomous Robots*.
- Supervised by Dr. Louise Dennis, Dr. Marie Farrell, and Prof. Michael Fisher.

University of Cambridge, BA (Hons) in Computer Science — Oct 2019 – Jun 2022

- Grade II.1 (70%).
- Dissertation titled *Delay-Tolerant Link-State Routing*. Implemented and empirically evaluated a failure-tolerant routing protocol in C to run on real Unix-based routers, comparing with traditional protocols. Supervised by Dr. Jackson Woodruff.

Research Interests

My interests lie in the intersection of autonomous robotics and formal methods. Specifically, I am passionate about verifying mission-critical systems under uncertainty and failure.

Professional Activities

- Part of the local organising committee of The 19th International Conference on Integrated Formal Methods 2024.
- Member of the Autonomy and Verification Network.

Teaching Experience

Graduate Teaching Assistant, University of Manchester — Sep 2024 – Present

- COMP21111 *Logic and Modelling*
- COMP11120 *Mathematical Techniques for Computer Science*

Supervisor (TA equiv.), University of Cambridge — Oct 2022 – Jun 2024

- CST IB *Logic and Proof*
- CST II *Hoare Logic and Model Checking*
- CST IB *Semantics of Programming Languages*

Industry Experience

Software Engineer, Tarides – Paris, France — Sep 2022 – Jun 2024

- Developed open-source CI (Continuous Integration) software for the OCaml language's ecosystem, involving systems programming to distribute dependency solving, compilation, and testing to many operating systems and architectures.
- Responsible for the CI of the Opam Repository, the central package universe for OCaml with tens of thousands of packages.

Embedded Systems Intern, Tarides – Paris, France — Jul 2021 – Sep 2021

- Research & development project to cross-compile the OCaml bytecode runtime to an ARM-based microcontroller using a real-time operating system as a base layer.

Computer Graphics Research Intern, University of Cambridge — Jun 2020 – Sep 2020

- Developed and implemented a visual model of ageing vision in virtual reality with Unity, as part of the Graphics and Displays Research Group.

Volunteering

Run Leader, Runwild – University of Manchester Dec 2024 – Present

- Lead bi-weekly group runs for a student running group.

English Teacher, Connections English Institute – Ibagué, Colombia Jul 2024 – Aug 2024

- Taught adult beginners in morning and night classes.